higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY

(First Paper)

NQF LEVEL 4

(10401034)

21 February 2019 (Y-Paper)
13:00–16:00

Non-programmable calculators and drawing instruments including rulers, pairs of compasses and protractors may be used.

This question paper consists of 11 pages and 1 answer sheet.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Clearly show ALL calculations, diagrams, graphs, etc. used to determine the answers.
5. Round off answers to TWO decimal places unless otherwise stated.
6. Use $\pi = 3,14$. Students will be penalised if any other value is used.
7. Answer QUESTION 4.1.5 on the ANSWER SHEET.
8. Diagrams are NOT necessarily drawn to scale.
9. Start each question on a NEW page.
10. Write neatly and legibly.

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QUESTION 1

1.1 Calculate the following without using a calculator. Show ALL steps followed to calculate the answer. Leave the answer as a fraction in its simplest form.

$$\frac{\sqrt{36}+\sqrt[3]{8}+2^3}{3(16+2)-22}$$ (4)

1.2 Rearrange the following decimal numbers in ascending order:

0,23; 0,2; -0,2; 0,201 (2)

1.3 Convert 500 000 grams to tonnes. Show ALL calculations.

(1 000 grams = 1 kilogram and 1 000 kilograms = 1 tonne) (4)

1.4 Extend the following pattern by writing down the next THREE numbers of the pattern:

13; 39; 117; …; …; … (3)

1.5 The price of a pair of shoes after an 8% increase is R1 296.

Calculate the price of the shoes before the increase. (4)

1.6 Sandy is in Singapore. Her friend from South Africa phones her at 16:58 South African time. Singapore time is 6 hours ahead of South Africa.

What time is it in Singapore when Sandy receives her friend's call? (2)

1.7 Calculate the petrol consumption of a car in km/ℓ if it uses 39,7 ℓ of petrol to cover a distance of 423 km. Round off the answer correctly to ONE decimal place. (3)

1.8 Convert 229 minutes to hours and minutes. (2)

1.9 Noah travels by taxi to a wedding venue that is 115 km away from his home. The taxi travels at an average speed of 127 km/h.

How long (in hours) will it take him to reach the wedding venue?

NOTE: Distance = speed $\times$ time (3)

1.10 Mary has R500, Alvin has R200 and Sipho has R300.

What is the ratio of the amount of money Sipho has to the amount of money Mary has to the amount of money Alvin has? Write the ratio in simplest form. (3)

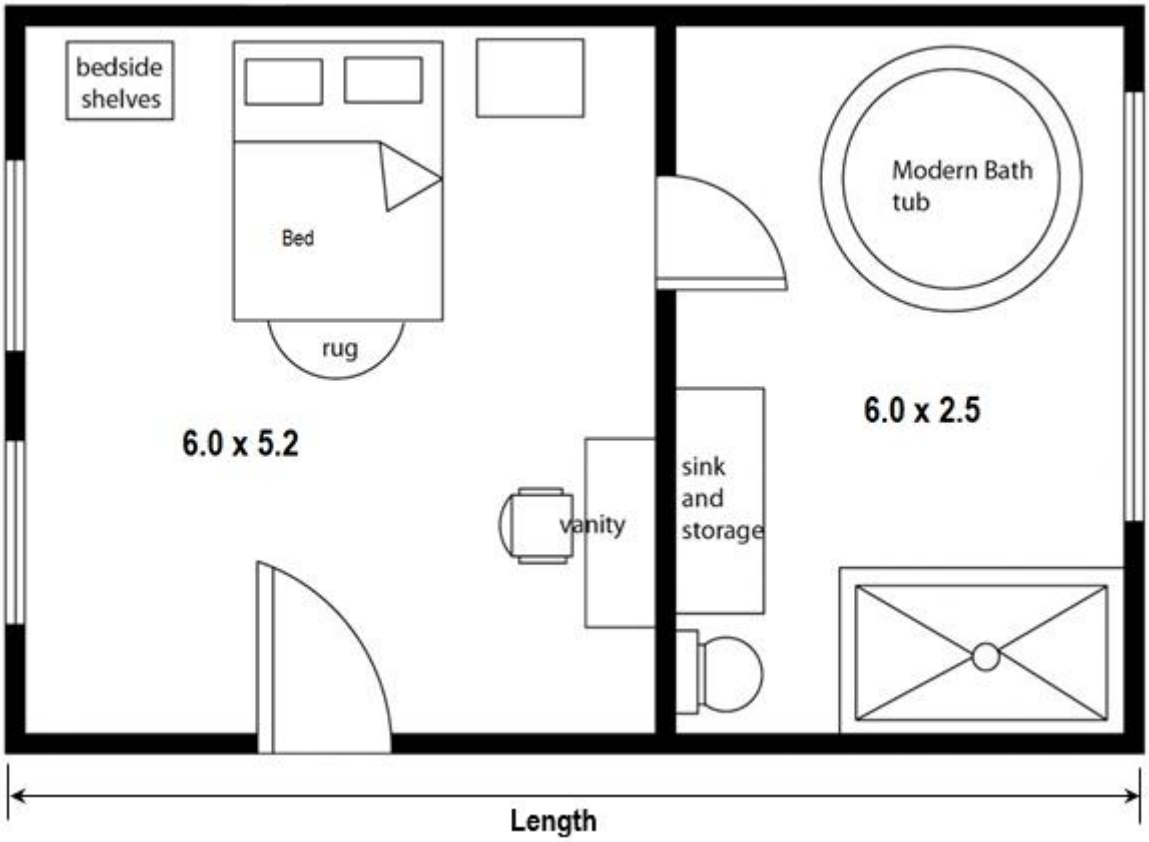
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QUESTION 2

2.1 John decided to build a house in an upmarket suburb for his family. The floor plan of a master bedroom and master bathroom of the house is shown below. The dimensions on the floor plan are in metres.

Study the floor plan and answer the questions.



2.1.1 How many windows are on the floor plan? (1)

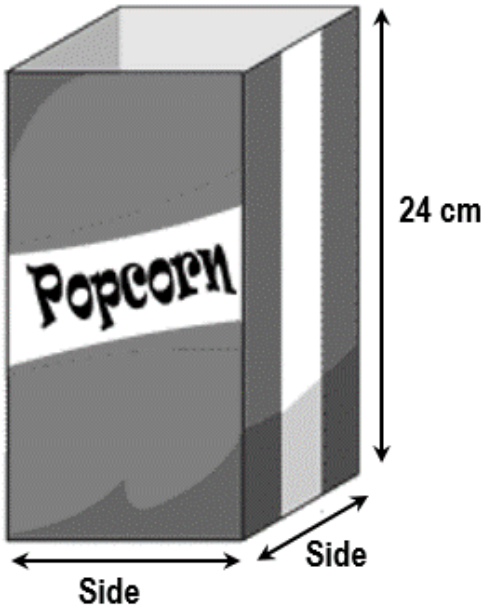
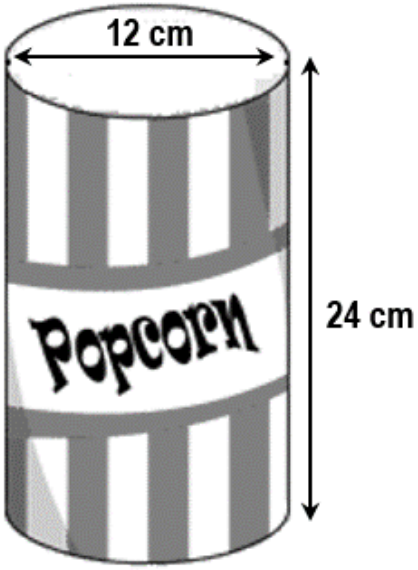
2.1.2 Use the dimensions on the floor plan to calculate the total area of the master bedroom and the master bathroom. (4)

2.1.3 Calculate the total length of the master bedroom and the master bathroom. (2)

2.2 The map of the suburb in which the house is built has a scale of 1:500 000. The map distance from the house to the nearest movie theatre is 15 mm.

Determine the actual distance (in kilometres) between the house and the movie theatre. (4)

2.3 John and his family went to the movie theatre. They decided to buy popcorn. The diagram below shows the different popcorn containers:

CONTAINER WITH SQUARE BASE	CYLINDRICAL CONTAINER
	
Volume of square-base container = 3 456 cm^3	Diameter of base = 12 cm

2.3.1 Determine the length of the side of the square-base container.

Formula: Volume = $side^2 \times height$ (5)

2.3.2 Calculate the radius of the cylindrical container. (2)

2.3.3 Calculate the difference between the volumes of the square-base container and the cylindrical container.

Formula: Volume of cylinder = $\pi \times r^2 \times height$ where $\pi = 3{,}14$ (6)

2.3.4 Calculate the total surface area of the cylindrical container.

Formula: Surface area = $\pi \times diameter \times h + \pi \times r^2$ where $\pi = 3{,}14$ (4)

2.3.5 What will the best unit of measure be to determine the mass of a container of popcorn? (2)

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QUESTION 3

3.1 Choose a definition from COLUMN B that matches a financial term in COLUMN A. Write only the letter (A–F) next to the question number (3.1.1–3.1.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1.1	VAT inclusive	A	total income from salaries before any deductions
3.1.2	PAYE	B	range of income subject to a particular tax rate
3.1.3	Gross salary	C	monthly deduction from salary paid towards income tax
3.1.4	Stock market	D	price that already includes tax
3.1.5	Tax bracket	E	place where shares of public-listed companies are traded
		F	price to which tax is yet to be added to arrive at the final cost

(5 × 1) (5)

3.2 John has a cheque account with Union Bank. The 2017 bank charges for Union Bank cheque-account holders are presented below:

Union Bank		
Monthly account fee		R110
Withdrawals:		Fees
Cash withdrawals @ Union Bank ATM	A combination of R4 500 free per month after which fees listed on the right apply	R1,55 per R100
Cash @ till		R1,80 per withdrawal
Other bank's ATM		R12 + R1,55 per R100
Cheque withdrawal at Union Bank		R55 + R1,75 per R100
Deposits:		
Cash deposit at Union Bank branch		R60 + R1,75 per R100
Cheque deposit at Union Bank branch and ATM		R55 + R7 per cheque

Study the given table to answer the following questions:

3.2.1 What is the monthly fee that John pays as a cheque account holder? (1)

3.2.2 John has issued a cheque for his assistant to withdraw R800 at a teller inside a Union Bank branch.

Calculate the bank charges on the R800 withdrawn. (2)

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3.2.3 John withdrew R700 from another bank's ATM.

Calculate the bank charges on the R700 he withdrew. (2)

3.2.4 The total cash that John withdrew from Union Bank ATMs during June was R6 350.

Calculate the total bank charges on the R6 350. (4)

3.2.5 John deposited R1 000 cash and a R5 000 cheque at a Union Bank branch.

Calculate the total bank charges on the deposits. (4)

3.3 Union Bank offers 14,5% compound interest per annum on an amount of money that is invested. John invested R20 000,00 for a period of 5 years. The table below is a statement of John's investment account.

Union Bank	**INTEREST CREDITED**	**BALANCE**
Opening balance		R20 000,00
End year 1	R2 900,00	R22 900,00
End year 2	R3 320,50	R26 220,50
End year 3	R3 801,97	R30 022,47
End year 4	R4 353,26	**A**
End year 5	**B**	**C**

3.3.1 Why is the interest earned by John at the end of year 2, not the same as the interest earned at the end of year 1? (1)

3.3.2 Determine the values of A, B and C. (6)

3.3.3 Calculate the interest John earned over the 5-year period. (2)

3.3.4 Calculate the interest John would have earned over the 5-year period if he had invested R20 000 at 14,5% simple interest per year. (3)

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QUESTION 4

John is planning a fundraising event for an orphanage. He must hire a venue to host the function. He obtained quotations from three different event venues. Each venue has a fixed rental cost and an additional cost per person.

TABLE 1 shows the costing structure of these three event venues:

VENUE	FIXED RENTAL COSTS	ADDITIONAL COST PER PERSON
Golden Sands	R2 500	R65
Fort Mistake	R5 000	R45
Castle Rock	R9 000	R25

TABLE 1

TABLE 2 represents the total cost in rand to hire each venue based on the number of people attending:

Number of people	0	50	100	150	200	250
Total cost at Golden Sands	**A**	5 750	9 000	12 250	15 500	18 750
Total cost at Fort Mistake	5 000	7 250	9 500	11 750	14 000	**B**
Total cost at Castle Rock	9 000	10 250	**C**	12 750	14 000	15 250

TABLE 2

Study the given tables to answer the following questions:

4.1 Use the information in the tables to extend the numerical patterns by completing the missing values A, B and C. Write only the answer next to the letter (A–C) in the ANSWER BOOK. (3)

4.2 Derive a formula to calculate the total cost to hire Fort Mistake. (3)

4.3 Use the formula in QUESTION 4.1.2 to calculate the maximum number of people that can attend the event if the total cost is R15 350. (4)

4.4 Determine the cheapest venue if only 120 people attend the fundraiser. Show ALL calculations. (4)

4.5 Use the tables to draw and label THREE line graphs on the ANSWER SHEET (attached) showing the total cost of each venue.

Label each line, label the horizontal and vertical axes and give the graph a suitable heading. (11)

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4.6 Use the line graphs in QUESTION 4.1.5 to determine the number of persons that can attend the function where the total cost of hiring Fort Mistake and Castle Rock will be the same. (2)

4.7 John sells the fundraiser tickets for R180 each.

Calculate the profit that he will make if he sells 200 tickets and hires the Castle Rock events venue. (3)

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QUESTION 5

The Pietermaritzburg Agency for Community Social Action (PACSA) released its report on the year-on-year inflation on a basket of goods. The table below shows the inflation prices of a sample of the PACSA basket of goods between January 2016 and January 2017.

Food grouping	Foods tracked	Quantity tracked	Jan 2016 price	Jan 2017 price	Year-on-year change (rand)	Year-on-year change (%)
Starchy foods	Maize meal	25 kg	R179,14	R244,32	R65,18	36%
	Rice	10 kg	R76,99	R80,82	R3,83	5%
	White bread	8 loaves	R84,84	R91,21	R6,37	8%
	Brown bread	4 loaves	R38,57	R41,94	R3,37	9%
Fat, oil	Cooking oil	4l	R87,14	R81,16	-R5,99	-7%
	Margarine	1 kg	R31,65	R36,48	R4,83	15%
Meat, eggs, fish	Eggs	30 eggs	R38,99	R45,32	**A**	16%
	Canned fish	4 cans	R59,29	R63,60	R4,31	7%
	Chicken pieces	6 kg	R131,97	R176,47	R44,50	34%
	Beef	1 kg	R58,33	R61,16	R2,83	5%
Vegetables	Carrots	2 kg	R16,97	R19,30	R2,33	14%
	Cabbage	2 heads	R28,55	R20,97	-R7,58	**B**
	Potatoes	10 kg	R73,32	R55,49	-R17,83	-24%
Total cost of PACSA food basket:			R905,75	R1 018,24	R112,49	12%

[Adapted from: *https://businesstech.co.za/news/lifestyle/155301/food-prices-in-south-africa-2016-vs-2017/*]

Study the given table to answer the following questions:

5.1 Define the term *inflation*. (2)

5.2 What is the percentage increase on the total cost of the food basket from January 2016 to January 2017? (1)

5.3 Calculate the value of A which represents the year-on-year change (in rand) on the price of eggs. (3)

5.4 Calculate the value of B which represents the year-on-year change (in percentage) on the price of TWO heads of cabbage. (3)

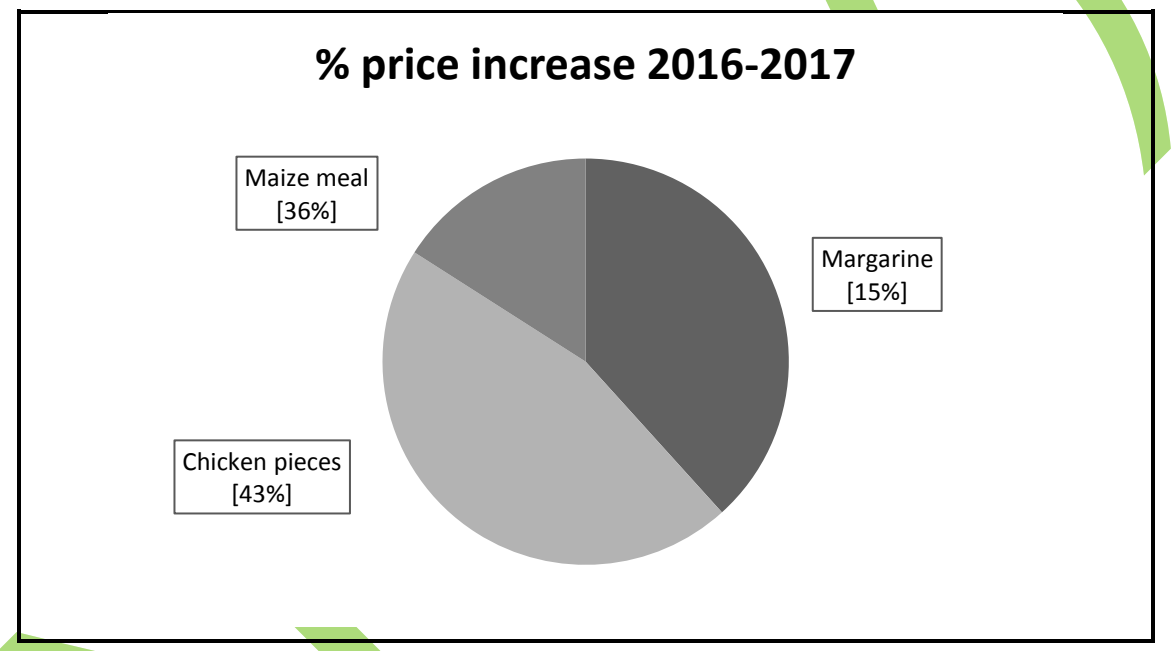
5.5 Calculate the median prices of the basket of food for January 2017. (4)

5.6 Calculate the range of the prices of the basket of food for January 2017. (3)

5.7 Calculate the difference between the mean (average) of the prices of the basket of food for January 2017 and January 2016. (6)

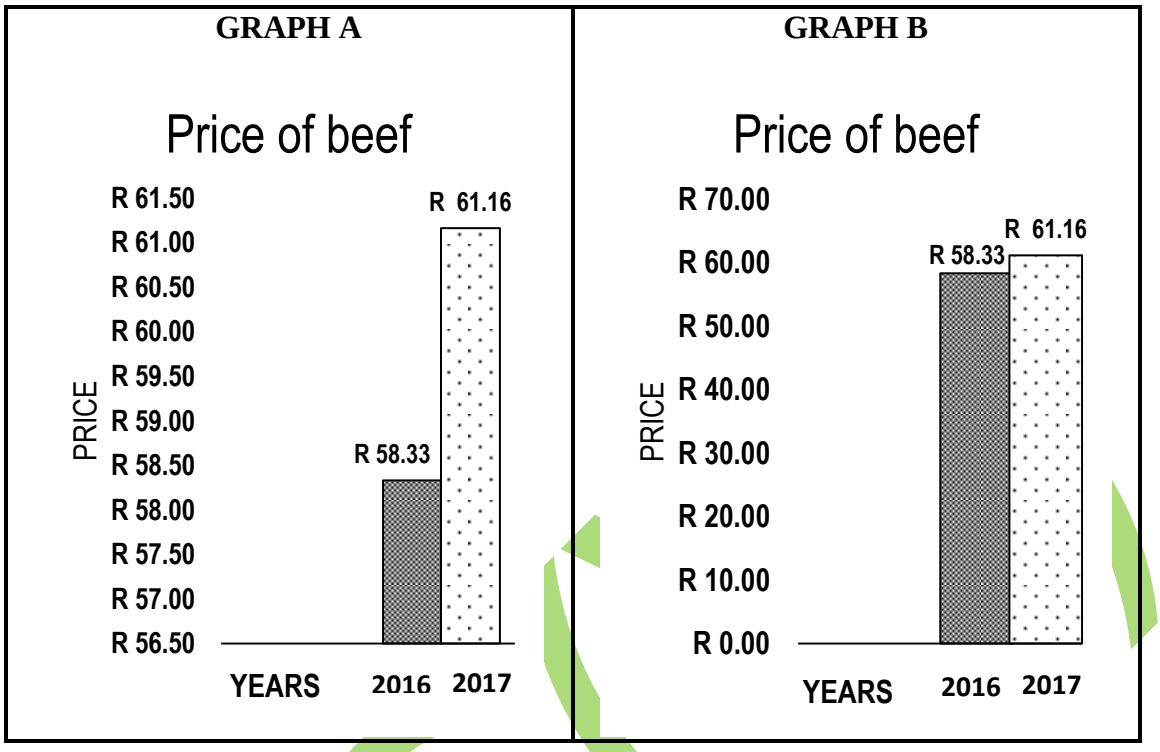
5.8 The pie chart below shows the percentage increase of maize meal, chicken pieces and margarine.

Identify the error in the pie chart.



(2)

5.9 Graph A and Graph B below show the change in the price of beef from 2016 to 2017:



Study the given graphs to answer the following questions:

5.9.1 What type of graph is represented above? (1)

5.9.2 Which of the two graphs is misleading? Give TWO reasons to substantiate the answer. (5)

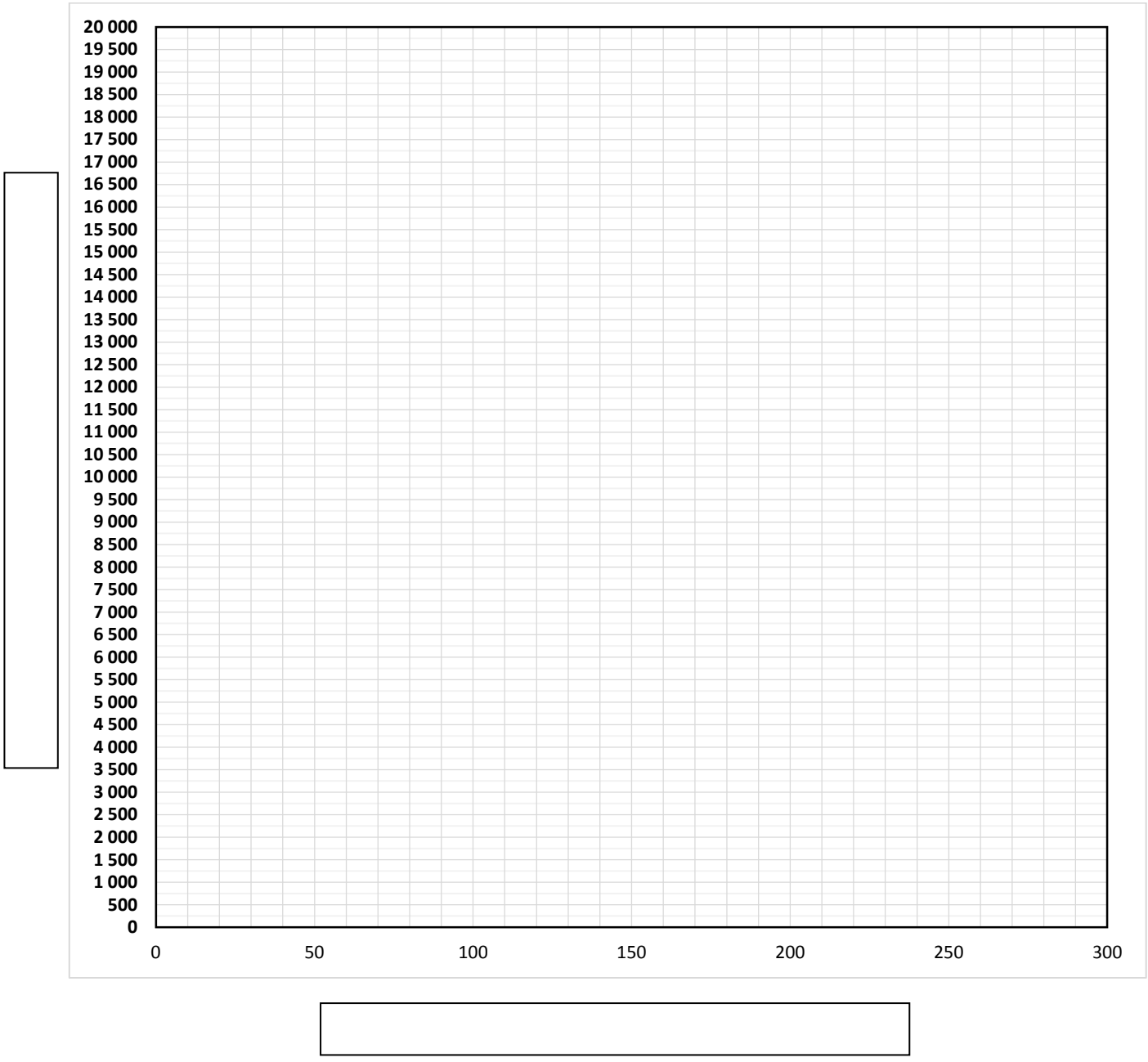
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TOTAL: 150

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ANSWER SHEET EXAMINATION NUMBER:

QUESTION 4.1.5



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